Journal of
Molluscan Studies

The Malacological Society of London

Journal of Molluscan Studies (2020) **0**: 1–7. doi:10.1093/mollus/eyaa020

Comparing shape along growth trajectories in two marine snail ecotypes of *Littorina saxatilis*: a test of evolution by paedomorphosis

J. Gefaell, N. Varela and E. Rolán-Alvarez

Facultade de Bioloxía e Centro de Investigación Mariña (CIM-UVIGO), Universidade de Vigo, E-36310 Vigo, Spain

Correspondence: E. Rolán-Alvarez; e-mail: rolan@uvigo.es

(Received 18 November 2019; editorial decision 25 May 2020)

ABSTRACT

Two sympatric ecotypes ('crab' and 'wave') of *Littorina saxatilis* are adapted to different microhabitats. It has been claimed, based on the comparison of proteomic differentiation across ontogeny, that the wave ecotype may have evolved by paedomorphosis from an ancestor more similar to the crab ecotype. Here, we test the paedomorphosis hypothesis at a morphological level by comparing crab and wave specimens from two localities using the pattern of shell shape differentiation across ontogeny. The results obtained show a pattern of shell shape differentiation similar to the one observed in previous proteomic studies, but such a pattern could be caused by different modes of evolution, and not necessarily by paedomorphosis. This study emphasizes that in addition to studying the pattern of differentiation, the direction of the evolutionary change across ontogeny has to be analysed before conclusions can be drawn on particular developmental modes of evolution.

INTRODUCTION

Today most historians of science agree that developmental biology was virtually absent from the Modern Synthesis (Grene & Depew, 2004; Amundson, 2005; Ruse, 2009; Pigliucci, 2017; but see Minelli, 2010). However, conceptual innovations and discoveries made throughout the second half of the 20th century (such as those related to homeotic genes and their conservation across the animal kingdom; Lewis, 1978; McGinnis *et al.*, 1984) prompted the foundation of a new discipline called evolutionary developmental biology (Raff, 1996; Hall, 1999, 2012; Carroll, 2005). Evolutionary developmental biology (also known as 'evo-devo') aims to understand the various ways in which development is related to evolution: how development evolves, how developmental plasticity contributes to the evolutionary origins of phenotypic traits or how modifications in developmental timing affect evolution (Hall, 2012). Today there seems to be a consensus among evolutionary biologists that a complete explanation of evolutionary processes requires taking into account the trajectories of ontogenetic development of organisms (e.g. Hall, 1999; Minelli & Fusco, 2012). Nevertheless, the exact relationship between evolution and development is not yet well understood in most traits and organisms.

Several hypotheses have been proposed to account for this relationship. Probably one of the most important is paedomorphosis, where evolutionary changes occur through the retention of ancestral juvenile traits in the adult form of the modern species (Gould, 1977). Paedomorphosis has been proposed as a mechanism to explain the evolution of certain groups of arthropods (McNamara, 1978), echinoderms (Stöhr & Martynov, 2016), nudibranchs (Korshunova *et al.*, 2018; Martynov *et al.*, 2020) and snails (Visser, 1988; Barnes, 1993; Ward & Slotow, 1997; Diz, Páez de la Cadena & Rolán-Alvarez, 2012). Furthermore, there is some evidence to suggest that, in the evolution of the human lineage, a process of neural paedomorphosis may have occurred (Somel *et al.*, 2009; Bufill, Agusti & Blesa, 2011).

Gould (1977) argued that paedomorphosis could occur through two different mechanisms: neoteny or progenesis. Neoteny involves the retention of juvenile traits through a slowing down of somatic development. As such, when the organism reaches the age of sexual maturity of its ancestor, it still presents phenotypic characteristics of a juvenile, while being a fully sexually mature organism. On the other hand, in the case of progenesis, paedomorphosis is achieved through the acceleration of sexual development in the modern species compared with its ancestor. Thus, the modern species is sexually mature at a much earlier stage than the ancestral species, and therefore its nonsexual phenotypic traits correspond to those of the ancestor's juvenile. Presumably, whether paedomorphosis is reached through neoteny or progenesis depends on the life history characteristics of the species, and on the particular molecular mechanisms that guide its ontogeny (Edelman, 1986). Consequently, its occurrence must be determined through molecular genetic techniques. Nonetheless, morphological studies should be carried out alongside molecular analyses, in order to elucidate the existence of any paedomorphosis-like trends at the phenotypic level in the development of the organism under scrutiny. Such an approach is necessary since it is usually at the morphological level where organisms ultimately show adaptations that affect their biological fitness and therefore could be favoured by natural selection

© *The Author(s) 2020. Published by Oxford University Press on behalf of The Malacological Society of London.*
All rights reserved. For permissions, please e-mail: journals.permissions@oup.com

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

(Diz *et al.*, 2012). Despite its potential importance, morphometric methods have been rarely used to evaluate the existence of paedomorphosis (but see Evin *et al.*, 2017). In this study, we perform morphological analyses in a putative case of paedomorphic evolution in a marine caenogastropod, the rough periwinkle *Littorina saxatilis* (Olivi, 1792).

This species has become one of the main models for the study of adaptation and speciation in sympatry through an ecological gradient (reviewed in Rolán-Alvarez, 2007; Rolán-Alvarez, Austin & Boulding, 2015). The rough periwinkle is distributed along the western coast of Europe, and its main ecological niche is the intertidal rocky shore, where it feeds on microalgae and diatoms (Reid, 1996). Taxonomic, ecological and genetic research has revealed that there are several ecotypes of this species, each one adapted to a different microhabitat (Johannesson, Johannesson & Rolán-Alvarez, 1993; Johannesson & Johannesson, 1996; Johannesson, Rolán-Alvarez & Erlandsson, 1997; Wilding, Butlin & Grahame 2001; Johannesson, 2003; Hollander, Lindegarth & Johannesson, 2005; Le Pennec *et al.*, 2017; review by Reid, 1996; Rolán-Alvarez *et al.*, 2015). This characteristic has been attributed to its low dispersal capability, having direct development and low mobility as adults (Reid, 1996), making it easier for new alleles to become fixed and leading to the species becoming adapted to the diverse environments it inhabits. These factors facilitate genetic differentiation, and therefore adaptation (Behrens Yamada, 1989).

Two different ecotypes have been described on rocky shores from northwest Spain (Rolán-Alvarez *et al.*, 2004; Butlin *et al.*, 2014; review by Rolán-Alvarez, 2007; Rolán-Alvarez *et al.*, 2015): one of them adapted to the lower shore ('wave' ecotype, previously 'smooth' and 'unbanded'), and the other adapted to the upper shore ('crab' ecotype, previously 'ridged' and 'banded'). The wave ecotype is small, with an average size about half of that of the crab ecotype. The former is also characterized by a thinner shell and larger shell aperture. These size and shape differences appear to represent adaptations to specific habitats of the shore. The wave ecotype's characteristics seem to be adaptations to the strong waves that impact the rocks on the lower shore it inhabits (Rolán-Alvarez, 2007; Rolán-Alvarez *et al.*, 2015). The larger shell aperture shown by the wave ecotype allows it to accommodate a larger foot, which in turn favours a better adhesion to substrate, preventing the individual from being dislodged by waves (Butlin *et al.*, 2014). The crab ecotype's characteristics seem to be adaptations to resist variable levels of temperature, desiccation and salinity, as well as greater prevalence of crab predation, typical of the upper regions of the shore (reviewed in Rolán-Alvarez, 2007).

These morphological trends show a strong genetic basis both within and between ecotypes (Wilding *et al.*, 2001; Conde-Padín *et al.*, 2007; Conde-Padín, Caballero & Rolán-Alvarez, 2009; Galindo *et al.*, 2019). The evolution of these ecotypes has involved a certain degree of gene flow through the occasional production of fertile intermediate progeny (Rolán-Alvarez, 2007). Previous studies suggest the existence of an ancestral variety from which the two current ecotypes evolved (Quesada *et al.*, 2007; Butlin *et al.*, 2014). Presumably, this ancestral variety would have been morphologically and ecologically similar to the crab ecotype, as it is now one of the most frequent phenotypes of the species (Johannesson & Johannesson, 1996; Wilding *et al.*, 2001; Galindo *et al.*, 2019; reviewed in Reid, 1996). Therefore, whether the wave ecotype has evolved from a crab ecotype-like ancestor remains unclear.

When performing a comparative proteomic analysis of these two ecotypes at different developmental stages, Diz *et al.* (2012) found a trend that suggested evolution by paedomorphosis. In particular, they provided evidence of a higher ontogenetic differentiation at the proteomic level in the crab ecotype (metamorphic) in comparison with the wave ecotype (paedomorphic), which suggests the possibility of juvenile stage retention in this latter ecotype. A comparison of venom proteins in the ontogeny of two related snake species found a similar proteomic trend, suggesting another case of evolution by paedomorphosis (Calvete *et al.*, 2010, 2011).

However, analyses of paedomorphic evolution are constrained when using only a proteomic approach, as the overall direction of change cannot be easily elucidated. Furthermore, as many of the adaptations of *L. saxatilis* ecotypes involve changes in shell morphology (reviewed in Rolán-Alvarez, 2007; Rolán-Alvarez *et al.*, 2015), the paedomorphic evolution hypothesis should be preferentially tested at the morphological level. Nevertheless, no morphological test of the paedomorphosis hypothesis in *L. saxatilis* ecotypes has been performed to date.

In this study, we evaluate the paedomorphic hypothesis for the evolution of the wave ecotype of *L. saxatilis* using geometric morphometric analyses of shell morphology. In particular, we test whether the shell characteristics of the wave ecotype evolved through a paedomorphic process from a crab-like ancestor. Our prediction is that, if the paedomorphic hypothesis is true, the shell shape of the adult wave ecotype should resemble that of the juvenile crab ecotype, to a greater extent than that of the adult crab ecotype. According to this hypothesis, the wave ecotype would have retained the juvenile traits of their crab-like ancestor during adulthood, therefore showing greater resemblance to the juvenile of the modern crab ecotype.

MATERIAL AND METHODS

Sampling

Littorina saxatilis specimens were collected from two sites in northwest Spain: Cabo Silleiro, on 15 October 2017, and La Cetarea, on 17 October 2018 (see Table 1 for further details). Within these localities, two subsamples were chosen to represent the two ecotypes, wave and crab. The crab subsample was taken from the barnacle belt (upper shore), whereas the wave ecotype was taken from the mussel belt (lower shore); these are the typical habitats of these ecotypes (Rolán-Alvarez *et al.*, 2015; review by Rolán-Alvarez, 2007). We sampled a few hundred specimens per locality and ecotype within an area of a few square metres. These specimens were then brought to the laboratory for morphometric study. Populations from Cabo Silleiro and La Cetarea show little genetic differentiation for several genetic markers (Rolán-Alvarez *et al.*, 2004; Galindo *et al.*, 2013) and were therefore considered replicates.

Experimental design and sample processing

The specimens were initially classified into five adult size classes in the laboratory based on the shell length (measured by Vernier callipers). The original idea was to have similar sample sizes per class following a stratified sampling design. However, specimens were reclassified *a posteriori* on the basis of the best geometric estimate of size (see below); the final specimen classification is shown in Table 1. In total, 188 individuals were analysed from Cabo Silleiro (90 crab and 98 wave specimens), and 133 were analysed from La Cetarea (68 crab and 65 wave specimens). The individuals were first photographed, then dissected and sexed (i.e. using the presence of penis or brood pouch). This species is ovoviviparous, so every adult female has a brood pouch with dozens of shelled embryos at different stages of development (Galindo *et al.*, 2019). Ten females from each ecotype and locality were randomly selected in order to take one fully developed embryo (juvenile stage) per female (40 embryos overall), so that the ontogenetic trajectory of the ecotypes could be studied. Specimens were photographed using a Leica digital ICA video camera mounted on a Leica MZ12 stereoscopic microscope following the methodology described by Conde-Padín *et al.* (2007). Images were taken using a 0.32 × lens (zoom 2.5) for the juvenile and adults, and a 1 × PLAN lens (zoom 2.75) for the embryos.

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

Table 1. Description of size classes used for the paedomorphosis and control tests in both ecotypes and localities.

			Size classes (centroid size)					
Locality	Ecotype	Variable	Class 0	Class 1	Class 2	Class 3	Class 4	Class 5
Cabo Silleiro	Crab	Mean ± SD	0.06 ± 0.005	0.32 ± 0.051	0.55 ± 0.048	0.77 ± 0.057	0.95 ± 0.067	1.19 ± 0.051
		Size range	0.052–0.066	0.211–0.428	0.463–0.627	0.685–0.866	0.877–1.074	1.133–1.308
		N	9	22	25	19	15	9
	Wave	Mean ± SD	0.07 ± 0.007	0.32 ± 0.036	0.44 ± 0.036	0.55 ± 0.038	0.67 ± 0.034	0.80 ± 0.066
		Size range	0.054–0.076	0.249–0.373	0.386–0.489	0.502–0.620	0.629–0.746	0.756–0.872
		N	10	32	16	32	15	3
La Cetarea	Crab	Mean ± SD	0.08 ± 0.005	0.46 ± 0.085	0.77 ± 0.083	1.07 ± 0.084	1.29 ± 0.023	1.68 ± 0.074
		Size range	0.073–0.088	0.300–0.599	0.611–0.892	0.934–1.200	1.250–1.321	1.615–1.812
		N	10	20	20	13	7	8
	Wave	Mean ± SD	0.08 ± 0.006	0.30 ± 0.043	0.49 ± 0.055	0.64 ± 0.038	0.89 ± 0.061	1.11 ± 0.028
		Size range	0.076–0.095	0.178–0.365	0.388–0.555	0.570–0.702	0.767–0.931	1.089–1.128
		N	10	16	20	21	6	2

The mean (±SD), range size and sample size (*n*) of the specimens analysed in each size class are shown.

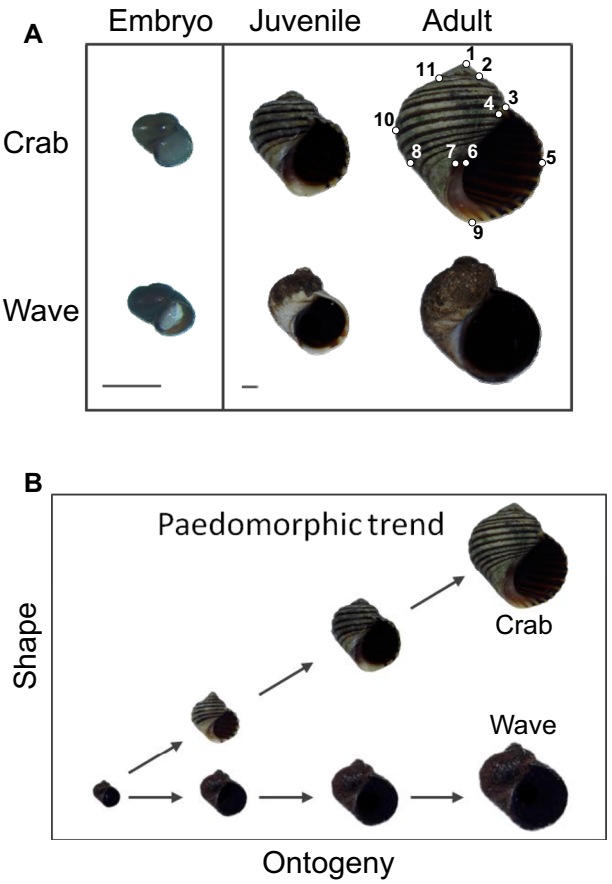


Figure 1. A. Experimental design, including embryos (age class 0) and adults (age classes 1–5). **B.** Logic of the paedomorphic trend: the hypothetical paedomorphic ecotype (wave) retains the shell shape of the juvenile crab ecotype throughout its ontogeny, whereas the shell shape of the crab ecotype keeps changing. This could be achieved by different developmental mechanisms (see the 'Results' section).

Morphometric analysis

Two-dimensional landmark (LM) coordinates were positioned on the digitized embryo and adult shells (Fig. 1A). This was accomplished using the TPS software suite (TpsUtil v.1.44 and TpsDig v.1.40; Rohlf, 2015), following the methodological procedure detailed elsewhere (Carvajal-Rodríguez, Conde-Padín & Rolán-Alvarez, 2005; Conde-Padín *et al.*, 2007, 2009; Galindo *et al.*, 2019). Shell size was defined as centroid size (CS), which is the square root of the sum of squared distances of the LMs to their centroid (Bookstein, 1991). Shell shape components were estimated by aligning raw specimen coordinates using the Procrustes generalized orthogonal method (Rohlf & Slice, 1990), which generates a reference configuration by minimizing squared distances between homologous LMs across specimens. Then, the partial warp scores (i.e. the linear projections of specimen deviations from the reference configuration in the multidimensional space of shape) are calculated. The principal components of the variation of the partial warp scores were finally obtained; these are referred to as relative warps (RWs; see Zelditch *et al.*, 2004). RW analysis was implemented in TpsRelw v.1.46 (Rohlf, 2015); this included the computation of the uniform component, following Galindo *et al.* (2019). A general and detailed description of this methodology is available in Zelditch *et al.* (2004: chapters 1–4). We obtained 18 RWs from the 11 LMs used, and the analyses were accomplished using only the first 11 RWs (RW_1–RW_{11}; these explained 95.4% of overall variation in shape). The results were summarized for just the first six RWs; RW_7–RW_{11} showed similar trends but are not presented here. The biological interpretation of the RWs was done by the thin-plate spline representation of different ecotypes and groups analysed (see below) using TpsRelw v.1.46. In these ecotypes, differences in RW_1 have been previously interpreted as differences in relative aperture area (Carvajal-Rodríguez *et al.*, 2005; Conde-Padín *et al.*, 2007, 2009; Galindo *et al.*, 2019). The relative aperture area was estimated *a posteriori* for the images obtained from the thin-plate spline representation with the program KLONK Image Measurement 15.1.1 (https://www.imagemeasurement.com/en/; Andersen, 1993), as a way to infer the true adaptive meaning of RW_1. One advantage of shell measurements is that they can be used to trace the direction of morphological change, either by using direct plots of measurements or by using the thin-plate spline representations of the first RW.

Statistical analysis

The minimum and maximum shell sizes were used to estimate the adult size range within ecotype and locality, and this range was subdivided into six size classes, five classes in the adult population (from 1 to 5) and one embryo class (0). Size classes are roughly correlated with snail age (Rolán-Alvarez *et al.*, 2015). Therefore, the relationship between shape and size class allows inference of changes in shape across growth trajectories.

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

Table 2. Paedomorphosis test for shell shape with the mean η^2 statistics obtained from one-way ANOVA (factor size class) within ecotype and locality (see the 'Material and Methods' section).

			Test		Control	
Experiment	Shape	Explained (%)	Crab	Wave	Crab	Wave
Shape	RW_1	26.6	0.41 ± 0.169	0.32 ± 0.182	0.50 ± 0.130	0.39 ± 0.447
	RW_2	20.5	0.20 ± 0.101	0.07 ± 0.018	0.06 ± 0.027	0.14 ± 0.011
	RW_3	16.3	0.79 ± 0.045	0.77 ± 0.039	0.86 ± 0.068	0.86 ± 0.022
	RW_4	9.2	0.41 ± 0.103	0.29 ± 0.238	0.06 ± 0.071	0.28 ± 0.379
	RW_5	6.9	0.32 ± 0.022	0.12 ± 0.051	0.38 ± 0.160	0.25 ± 0.111
	RW_6	4.6	0.37 ± 0.260	0.10 ± 0.002	0.44 ± 0.328	0.27 ± 0.013
	RW_7	3.3	0.33 ± 0.046	0.15 ± 0.088	0.10 ± 0.147	0.15 ± 0.148
	RW_8	2.7	0.07 ± 0.073	0.11 ± 0.048	0.08 ± 0.048	0.15 ± 0.088
	RW_9	2.2	0.17 ± 0.115	0.18 ± 0.142	0.10 ± 0.147	0.26 ± 0.095
	RW_{10}	1.8	0.21 ± 0.203	0.06 ± 0.033	0.09 ± 0.113	0.12 ± 0.057
	RW_{11}	1.6	0.26 ± 0.210	0.09 ± 0.019	0.30 ± 0.413	0.13 ± 0.006
	Average ± SD		0.26 ± 0.201	0.10 ± 0.078	0.27 ± 0.284	0.27 ± 0.249
	Binomial test (Cabo Silleiro)		0.0538 (9:2)		0.1612 (3:8)	
	Binomial test (La Cetarea)		0.0538 (9:2)		0.5468 (7:4)	
	Binomial test (both localities)		0.0034 (18:4)		0. 3084 (10:12)	
Proteomics (source: Diz *et al.*, 2012)	452 spots		0.53	0.46	0.46	0.45
	P		0.0066		0.1023	

In the paedomorphic test subset of data, all age classes were included in both ecotypes, while in the paedomorphic control subset of data, different age classes were compared between ecotypes (embryo and largest adult size for the wave ecotype; embryo and smallest adult size for the crab ecotype). For a significant paedomorphic pattern to be observed, the results for the paedomorphic and paedomorphic control tests should be significant and nonsignificant, respectively (see the 'Material and Methods' section). In binomial tests, we examined the null hypothesis that there was no bias in the values for η^2 in comparisons between crab and wave ecotypes for populations at the two locations separately, and together. The number of positive and negative comparisons across traits is shown in parentheses.

To quantify the degree of variation across the growth trajectory within ecotype and locality, we used one-way ANOVA, with the shell shape as the dependent variable and size class as a fixed factor. This was done in order to directly compare putative paedomorphic patterns of shell shape with proteomic data (see Diz *et al.*, 2012; Table 1). The factor-effect size (portion of dependent variable explained by the factor) was estimated using the eta squared coefficient (η^2) (Sokal & Rohlf, 1995). We compared ecotypes for paedomorphic test and paedomorphic control using different subsets of data and η^2. The paedomorphic test dataset included data from the six size classes of both ecotypes. The paedomorphic control dataset was a subset of the paedomorphic test dataset, which included only distinct size classes per ecotype: crab (sizes 0 and 1, representing embryos and the smallest adult or juvenile size class, respectively) and wave (sizes 0 and 4–5, representing embryos and the largest adult classes, respectively). Under a paedomorphic scenario, a significant paedomorphic test (but not a significant paedomorphic control) would be predicted because the shell morphology of the adult wave ecotype (size classes 4–5) should resemble the juvenile stage of the crab ecotype (size class 1). Analyses were carried out for each locality separately and as a single pooled dataset. Deviations from normality were tested by the Kolmogorov–Smirnov test (Sokal & Rohlf, 1995). The statistical analyses were carried out in SPSS v.24 (IBM Corp., USA) using the general linear model function.

RESULTS

The percentage of variance explained by each RW is presented in Table 2. None of the variables (CS and RWs) within ecotype, size class and locality deviated significantly from normality after multiple-comparison adjustment. Due to the stratified sampling, we explicitly checked whether the variation for different variables across size classes was similar by comparing the paedomorphic test and the paedomorphic control using η^2 between ecotypes (Table 2). Consistent with the trend previously observed using a proteomic approach, the paedomorphic test was significant while the paedomorphic control was not (compare probabilities for proteomics and binomial pooled as shown in Table 2). Moreover, the difference in mean η^2 between ecotypes in the paedomorphic test was double that recorded for the paedomorphic control, suggesting a paedomorphic trend. These results were qualitatively and quantitatively similar for each locality (Table 2).

The analyses of the morphometric trends, however, do not fit any paedomorphic pattern. For example, the trait lines of both ecotypes from Figure 2 do not resemble the trend outlined in Figure 1B. Furthermore, we did not observe any paedomorphic trend when the change in the RW_1 trait, the relative shell aperture, was compared between ecotypes across the ontogeny (Fig. 3).

DISCUSSION

We investigated whether the evolution of the wave ecotype of *Littorina saxatilis*, presumably derived from a crab-like ancestor (Diz *et al.*, 2012), occurred through a process of paedomorphosis. This hypothesis was proposed based on the higher degree of proteomic differentiation across ontogeny in the wave ecotype compared to the crab ecotype (Diz *et al.*, 2012). The same argument has been used to invoke evolution by paedomorphosis in other organisms (Calvete *et al.*, 2010, 2011). In this study, we explored paedomorphosis in *L. saxatilis* by analysing shell shape, in order to determine whether evolution by paedomorphosis is contributing to the known adaptive divergence in shell shape differences between these ecotypes (Johannesson *et al.*, 1993, 1997; Johannesson & Johannesson, 1996; Wilding *et al.*, 2001; Hollander *et al.*, 2005; Le Pennec *et al.*, 2017; review by Johannesson, 2003; Rolán-Alvarez *et al.*, 2015). Evolution by paedomorphosis in this species would be evidenced by the greater resemblance in shell shape between adult wave and juvenile crab ecotypes, than between the adults of both ecotypes. In particular, and following the strategy outlined by Diz *et al.* (2012), we predicted differences between ecotypes in shape differentiation

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

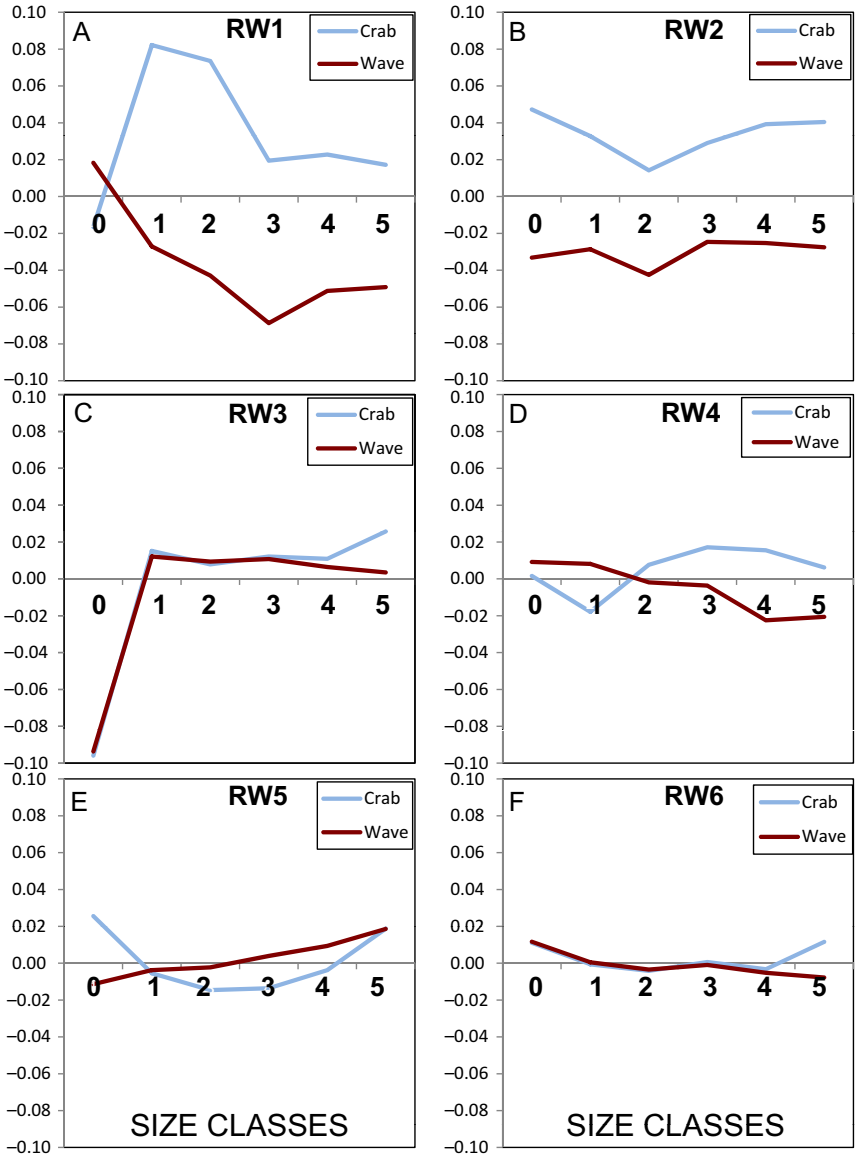


Figure 2. A–F. Growth trajectories of the six main relative warps for both ecotypes (values averaged between localities).

across ontogeny in the paedomorphic test but not in the paedomorphic control subset.

The quantitative analysis showing the amount of divergence in shell shape across ontogeny for the paedomorphic test and the paedomorphic control yielded results equivalent to those observed by Diz *et al.* (2012) using a proteomic approach. While the result of such test was considered in the past as an evidence for a paedomorphic trend, this conclusion is, however, premature. This is because when testing a paedomorphic hypothesis it is important to distinguish between what we have called the pattern of change and the direction of change of the phenotypic trait(s) under analysis. Although the two concepts may greatly resemble each other, they do not have the same properties.

The pattern of change refers to the magnitude of change of a given trait across a certain period of time, as well as to how this change varies in intensity across that period of time. In contrast, the direction of change refers to the particular modifications of the phenotype that accompany that change. In this sense, the direction of change of a particular phenotypic trait bears great resemblance to a change in the position of the animal in a particular morphospace after that change has occurred. A morphospace (Ridley, 2004; Brakefield, 2014) is an *N*-dimensional geometric representation of all the possible forms that a particular animal can pass through during their ontogeny and/or evolution. This distinction between pattern and direction of change is crucial because two organisms with the same pattern of change across ontogeny could experience opposing directions of change, ending at opposite ends of the morphospace.

Shell shape analysis allows the study of the direction of change across development. One way of achieving this is by using a thin-plate spline representation, which permits a straightforward morphological interpretation of the pattern of change observed through statistical analysis of the shell shape (Zelditch *et al.*, 2004: 129–152). Using shell shape data for wave and crab ecotypes, we found that the direction of change in shape across ontogeny for the six main shape variables (RW_1–RW_6) and the thin-plate representation of the shell shape aperture did not support a paedomorphic mode of evolution. In particular, shell shape change across ontogeny did not lead to greater resemblance between the juvenile crab ecotype and the adult wave ecotype. In summary, although

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

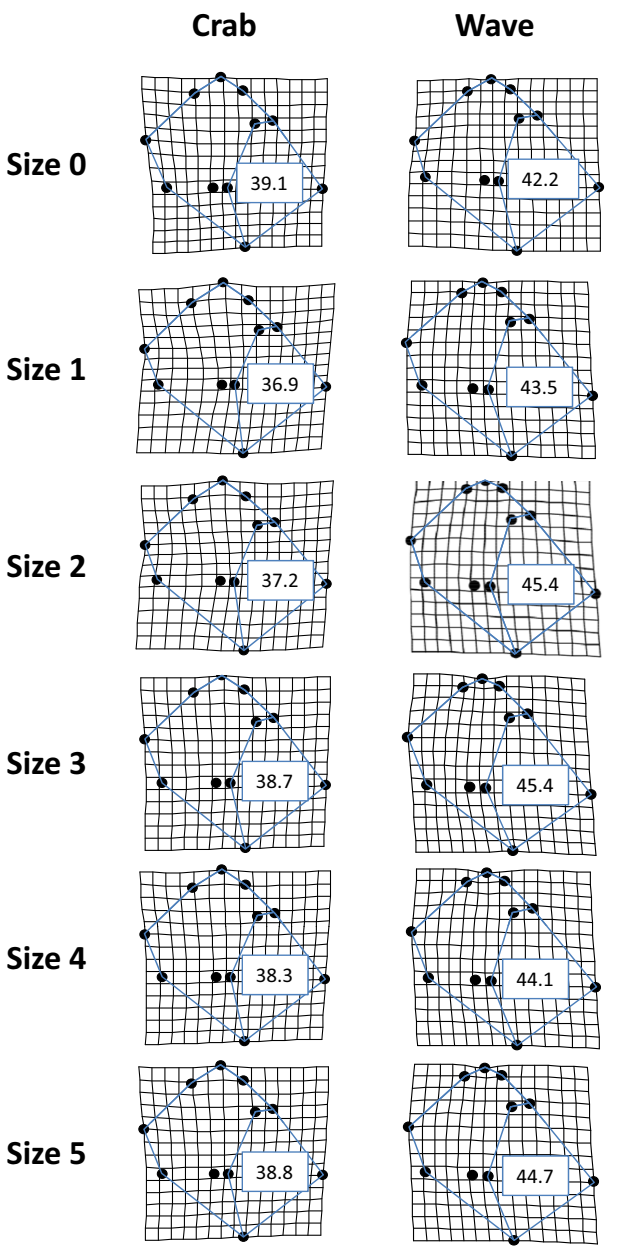


Figure 3. Thin-plate spline representation of first relative warp (RW_1) for the different size classes and ecotypes. After the external landmarks and those defining the aperture were connected, the relative proportion of the apertural area compared to the overall shell area was calculated using the software KLONK Image Measurement (see the 'Material and Methods' section).

the pattern of change of shell shape for the two ecotypes was compatible with paedomorphosis, the direction of change observed did not support this hypothesis, as the changes in the morphospace did not fit the predicted trajectory. This result calls for caution when inferring the existence of a paedomorphic trend on the basis of proteomic analysis alone. This is because while the test developed and used by Diz *et al.* (2012) fulfils the necessary condition for a process like paedomorphosis, on its own it is not sufficient to prove it.

Nevertheless, on the basis of our findings, we cannot discard the paedomorphic hypothesis as an explanation for the evolution of the wave ecotype. This is because of the limitations in the experimental procedures used by us. First, the sample size in a few classes was low, and this could account for the observed trends. Second, and more importantly, early embryonic development was not considered in our study, the embryos used all being fully grown ones. It is possible that paedomorphosis only affects the early stages of ontogenic development of the wave and crab ecotypes, so that fully grown embryos, juveniles and adults of different sizes do not develop in a paedomorphic-like trend. Morphological analyses that include early embryonic development are needed.

Notwithstanding these caveats, it is important to test evo-devo hypotheses, such as paedomorphosis. The Modern Synthesis, even with all its successes, has tended to focus on natural selection at the population level (i.e. in selecting the fittest genes within a given population), neglecting the mechanisms by which the fittest phenotypes arise in the first place. Evolutionary developmental biology, with its emphasis on patterns and directions of change in morphological traits and the regulatory gene changes underpinning them, could fill a gap in evolutionary studies. As highlighted by Gilbert (2003), filling this gap would shed light on the issue of the arrival of the fittest, rather than purely on the survival of the fittest.

ACKNOWLEDGEMENTS

We thank Mary Riádigos for administrative assistance, and Dr Manuela Truebano from Plymouth University and two anonymous referees for revising a later version of the manuscript. This work was supported by Xunta de Galicia (grant no. ED431C 2016-037), FONDOS FEDER ('Unha maneira de facer Europa') and the Ministerio de Economía, Industria y Competitividad (grant no. CGL2016-75482-P).

REFERENCES

AMUNDSON, R. 2005. *The changing role of the embryo in evolutionary thought.* Cambridge University Press, Cambridge.

ANDERSEN, C. 1993. KLONK Image Measurement. Available at: https://www.imagemeasurement.com/. Accessed 1 November 2018.

BARNES, R.S.K. 1993. On the nature of the coastal lagoon winkles attributed to *Littorina tenebrosa* and *Littorina saxatilis*. *Cahiers de Biologie Marine*, **34**: 477–495.

BEHRENS YAMADA, S. 1989. Are direct developers more locally adapted than planktonic developers? *Marine Biology*, **103**: 303–411.

BOOKSTEIN, F.L. 1991. *Morphometric tools for landmark data.* Cambridge University Press, Cambridge.

BRAKEFIELD, P.M. 2014. Evolution and development: organisms. In: *The Princeton guide to evolution* (J.B. Losos, ed.), pp. 436–443. Princeton University Press, Princeton, NJ.

BUFILL, E., AGUSTI, J. & BLESA, R. 2011. Human neoteny revisited: the case of synaptic plasticity. *American Journal of Human Biology*, **23**: 729–739.

BUTLIN, R.K., SAURA, M., CHARRIER, G., JACKSON, B., ANDRÉ, C., CABALLERO, A., COYNE, J.A., GALINDO, J., GRAHAME, J.W., HOLLANDER, J., KEMPPAINEN, P., MARTÍNEZ-FERNÁNDEZ, M., PANOVA, M., QUESADA, H., JOHANNESSON, K. & ROLÁN-ALVAREZ, E. 2014. Parallel evolution of local adaptation and reproductive isolation in the face of gene flow. *Evolution*, **68**: 935–949.

CALVETE, J.J., SANZ, L., CID, P., De La TORRE, P., FLORES-DÍAZ, M., DOS SANTOS, M.C., BORGES, A., BREMO, A., ANGULO, Y., LOMONTE, B., ALAPE-GIRÓN, A. & GUTIÉRREZ, J.M. 2010. Snake venomics of the central American rattlesnake *Crotalus simus* and the South American *Crotalus durissus* complex points to neurotoxicity as an adaptive paedomorphic trend along *Crotalus* dispersal in South America. *Journal of Proteome Research*, **9**: 528–544.

CALVETE, J.J., SANZ, L., PÉREZ, A., BORGES, A., VARGAS, A.M., LOMONTE, B., ANGULO, Y., GUTIÉRREZ, J.M., CHALKIDIS, H.M., MOURÃO, R.H., FURTADO, M.F. & MOURA-Da-SILVA, A.M. 2011. Snake population venomics and antivenomics of *Bothrops atrox*: paedomorphism along its transamazonian dispersal and implications of geographic venom variability on snakebite management. *Journal of Proteomics*, **74**: 510–527.

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020

CARROLL, S.B. 2005. *Endless forms most beautiful: the new science of evo-devo and the making of the animal kingdom*. W.W. Norton & Company, New York.

CARVAJAL-RODRÍGUEZ, A., CONDE-PADÍN, P. & ROLÁN-ALVAREZ, E. 2005. Decomposing shell form into size and shape by geometric morphometric methods in two sympatric ecotypes of *Littorina saxatilis*. *Journal of Molluscan Studies*, **71**: 313–318.

CONDE-PADÍN, P., CABALLERO, A. & ROLÁN-ALVAREZ, E. 2009. Relative role of gene determination and plastic response during ontogeny for shell shape traits subjected to diversifying selection. *Evolution*, **63**: 1356–1363.

CONDE-PADÍN, P., CARVAJAL-RODRÍGUEZ, A., CARBALLO, M., CABALLERO, A. & ROLÁN-ALVAREZ, E. 2007. Genetic variation for shell traits in a direct-developing marine snail involved in a putative sympatric ecological speciation process. *Evolutionary Ecology*, **21**: 635–650.

DIZ, A.P., PÁEZ De La CADENA, M. & ROLÁN-ALVAREZ, E. 2012. Proteomic evidence of a paedomorphic evolutionary process within a marine snail species: a strategy for adapting to extreme ecological conditions? *Journal of Evolutionary Biology*, **25**: 2569–2581.

EDELMAN, G.M. 1986. Molecular mechanisms of morphologic evolution. In: *Molecular evolution of life* (H. Baltscheffsky,H. Jörnvall & R. Rigler, eds), pp. 363–375. Cambridge University Press, Cambridge.

EVIN, A., OWEN, J., LARSON, G., DEBIAIS-THIBAUD, M., CUCCHI, T., VIDARSDOTTIR, U.S. & DOBNEY, K. 2017. A test for paedomorphism in domestic pig cranial morphology. *Biology Letters*, **13**: 20170321.

GALINDO, J., CACHEDA, D., CABALLERO, A. & ROLÁN-ALVAREZ, E. 2019. Untangling the contribution of genetic and environmental effects to shell differentiation across an environmental cline in a marine snail. *Journal of Experimental Marine Biology and Ecology*, **513**: 27–34.

GALINDO, J., MARTÍNEZ-FERNÁNDEZ, M., RODRÍGUEZ-RAMILLO, S.T. & ROLÁN-ALVAREZ, E. 2013. The role of local ecology during hybridization at the initial stages of ecological speciation in a marine snail. *Journal of Evolutionary Biology*, **26**: 1472–1487.

GILBERT, S.F. 2003. *Developmental biology*. Edn 7. Sinauer Associates, Sunderland, MA.

GRENE, M. & DEPEW, D. 2004. *The philosophy of biology: an episodic history*. Cambridge University Press, Cambridge.

GOULD, S.J. 1977. *Ontogeny and phylogeny*. Belknap Harvard, Cambridge, MA.

HALL, B.K. 1999. *Evolutionary developmental biology*. Edn 2. Kluwer Academic Publishers, London.

HALL, B.K. 2012. Evolutionary developmental biology (evo-devo): past, present, and future. *Evolution Education and Outreach*, **5**: 184–193.

HOLLANDER, J., LINDEGARTH, M. & JOHANNESSON, K. 2005. Local adaptation but not geographical separation promotes assortative mating in a snail. *Animal Behaviour*, **70**: 1209–1219.

JOHANNESSON, B. & JOHANNESSON, K. 1996. Population differences in behaviour and morphology in the snail *Littorina saxatilis*: phenotypic plasticity or genetic differentiation? *Journal of Zoology*, London, **240**: 475–493.

JOHANNESSON, K. 2003. Evolution in *Littorina*: ecology matters. *Journal of Sea Research*, **49**: 107–117.

JOHANNESSON, K., JOHANNESSON, B. & ROLÁN-ALVAREZ, E. 1993. Morphological differentiation and genetic cohesiveness over a microenvironmental gradient in the marine snail *Littorina saxatilis*. *Evolution*, **47**: 1770–1787.

JOHANNESSON, K., ROLÁN-ALVAREZ, E. & ERLANDSSON, J. 1997. Growth rate differences between upper and lower shore ecotypes of the marine snail *Littorina saxatilis* (Olivi) (Gastropoda). *Biological Journal of the Linnean Society*, **61**: 267–279.

KORSHUNOVA, T., LUNDIN, K., MALMBERG, K., PICTON, B. & MARTYNOV, A. 2018. First true brackish-water nudibranch mollusc provides new insights for phylogeny and biogeography and reveals paedomorphosis-driven evolution. *PLoS One*, **13**: e0192177.

Le PENNEC, G., BUTLIN, R.K., JONSSON, P. & JOHANNESSON, K. 2017. Adaptation to dislodgement risk on wave-swept rocky shores in the snail *Littorina saxatilis*. *PLoS One*, **12**: e0186901.

LEWIS, E.B. 1978. A gene complex controlling segmentation in *Drosophila*. *Nature*, **276**: 565–570.

McGINNIS, W., GARBER, R.L., WIRZ, J., KUROIWA, A. & GEHRING, W.J. 1984. A homologous protein-coding sequence in *Drosophila* homoeotic genes and its conservation in other metazoans. *Cell*, **37**: 403–408.

McNAMARA, K.J. 1978. Paedomorphosis in Scottish olenellid trilobites (early Cambrian). *Palaeontology*, **21**: 635–655.

MARTYNOV, A., LUNDIN, K., PICTON, B., FLETCHER, K., MALMBERG, K. & KORSHUNOVA, T. 2020. Multiple paedomorphic lineages of soft-substrate burrowing invertebrates: parallels in the origin of *Xenocratena* and *Xenoturbella*. *PLoS One*, **15**: e0227173.

MINELLI, A. 2010. Evolutionary developmental biology does not offer a significant challenge to the neo-Darwinian paradigm. In: *Contemporary debates in philosophy of biology* (F.J. Ayala & R. Arp, eds), pp. 213–226. Wiley–Blackwell, Oxford.

MINELLI, A. & FUSCO, G. 2012. *Evolving pathways: key themes in evolutionary developmental biology*. Cambridge University Press, Cambridge.

PIGLIUCCI, M. 2017. Darwinism after the modern synthesis. In: *The Darwinian tradition in context: research programs in evolutionary biology* (R.G. Delisle, ed.), pp. 89–103. Springer International Publishing, Cham, Switzerland.

QUESADA, H., POSADA, D., CABALLERO, A., MORÁN, P. & ROLÁN-ALVAREZ, E. 2007. Phylogenetic evidence for multiple sympatric ecological diversification in a marine snail. *Evolution*, **61**: 1600–1612

RAFF, R.A. 1996. *The shape of life: genes, development, and the evolution of animal form*. Chicago University Press, Chicago, IL.

REID, D.G. 1996. *Systematics and evolution of Littorina*. Ray Society, Andover, UK.

RIDLEY, M. 2004. *Evolution*. Edn 3. Blackwell Publishing, Oxford.

ROHLF, F.J. 2015. The tps series of software. *Hystrix*, **26**: 9–12.

ROHLF, F.J. & SLICE, D. 1990. Extensions of the Procrustes method for the optimal superimposition of landmarks. *Systematic Zoology*, **39**: 40–59.

ROLÁN-ALVAREZ, E. 2007. Sympatric speciation as a by-product of ecological adaptation in the Galician *Littorina saxatilis* hybrid zone. *Journal of Molluscan Studies*, **73**: 1–10.

ROLÁN-ALVAREZ, E., AUSTIN, C.J. & BOULDING, E.G. 2015. The contribution of the genus *Littorina* to the field of evolutionary ecology. *Oceanography and Marine Biology*, **53**: 157–214.

ROLÁN-ALVAREZ, E., CARBALLO, M., GALINDO, J., MORÁN, P., FERNÁNDEZ, B., CABALLERO, A., CRUZ, R., BOULDING, E. & JOHANNESSON, K. 2004. Nonallopatric and parallel origin of local reproductive barriers between two snail ecotypes. *Molecular Ecology*, **13**: 3415–3424.

RUSE, M. 2009. The history of evolutionary thought. In: *Evolution: the first four billion years* (M. Ruse & J. Travis, eds), pp. 67–86. Harvard University Press, Cambridge, MA.

SOKAL, R.R & ROHLF, F.J. 1995. *Biometry*. Edn 3. W.H. Freeman & Co., New York.

SOMEL, M., FRANZ, H., YAN, Z., LORENC, A., GUO, S., GIGER, T., KELSO, J., NICKEL, B., DANNEMANN, M., BAHN, S., WEBSTER, M.J., WEICKERT, C.S., LACHMANN, M., PÄÄBO, S. & KHAITOVICH, P. 2009. Transcriptional neoteny in the human brain. *Proceedings of the National Academy of Sciences of the United States of America*, **106**: 5743–5748.

STÖHR, S. & MARTYNOV, A. 2016. Paedomorphosis as an evolutionary driving force: insights from deep-sea brittle stars. *PLoS One*, **11**: e0164562.

VISSER, M.H.C. 1988. The significance of terminal duct structures and the role of neoteny in the evolution of the reproductive system of Pulmonata. *Zoologica Scripta*, **17**: 239–252.

WARD, D. & SLOTOW, R. 1997. Paedomorphosis and shell-colour polymorphism of a desert landsnail. *Journal of Arid Environments*, **36**: 677–685.

WILDING, C.S., BUTLIN, R.K. & GRAHAME, J. 2001. Differential gene exchange between parapatric morphs of *Littorina saxatilis* detected using AFLP markers. *Journal of Evolutionary Biology* **14**: 611–619.

ZELDITCH, M.L., SWIDERSKI, D.L., SHEETS, H.D. & FINK, W.L. 2004. *Geometric morphometrics for biologists: a primer*. Elsevier/Academic Press, Amsterdam.

Downloaded from https://academic.oup.com/mollus/advance-article/doi/10.1093/mollus/eyaa020/5899079 by guest on 31 August 2020